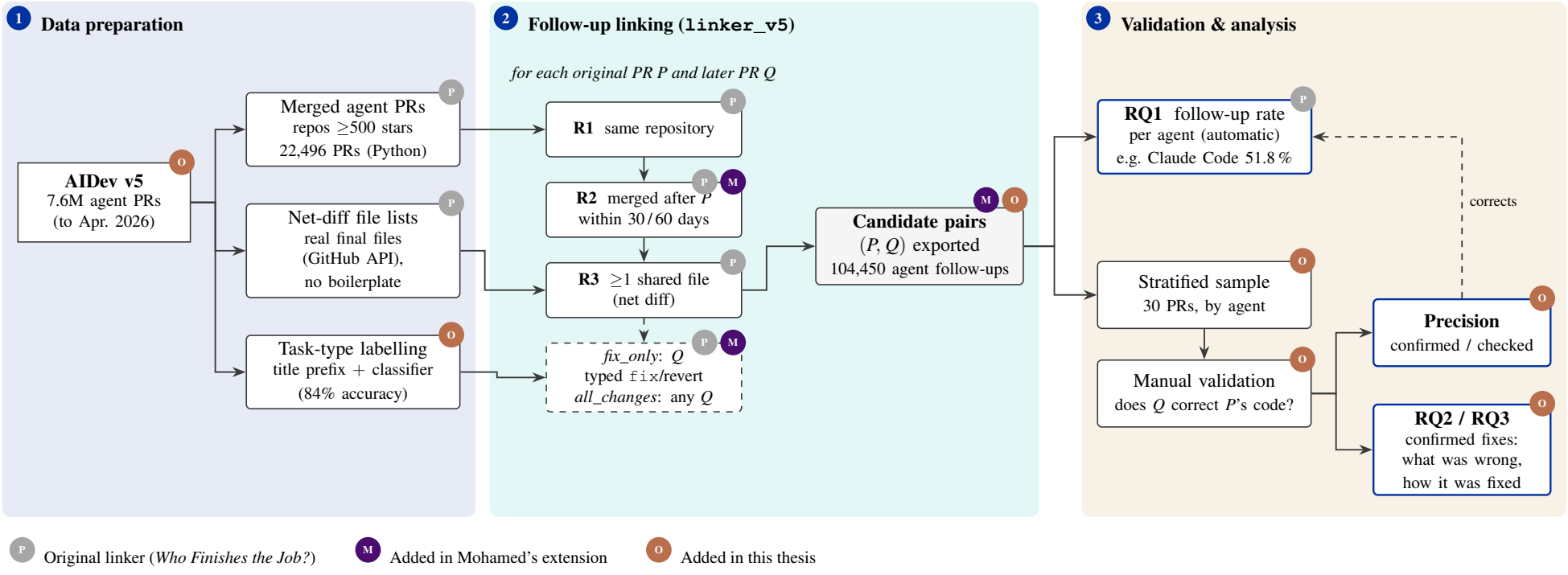

1 Data preparation
AIDev v5
7.6M agent PRs
(to Apr. 2026)
Merged agent PRs
repos ≥500 stars
22,496 PRs (Python)
Net-diff file lists
real final files
(GitHub API),
no boilerplate
Task-type labelling
title prefix + classifier
(84% accuracy)
2 Follow-up linking (linker_v5)
for each original PR P and later PR Q
R1 same repository
R2 merged after P
within 30/60 days
R3 ≥1 shared file
(net diff)
fix_only: Q
typed fix/revert
all_changes: any Q
Candidate pairs
(P, Q) exported
104,450 agent follow-ups
3 Validation & analysis
RQ1 follow-up rate
per agent (automatic)
e.g. Claude Code 51.8 %
corrects
Stratified sample
30 PRs, by agent
Manual validation
does Q correct P's code?
Precision
confirmed / checked
RQ2 / RQ3
confirmed fixes:
what was wrong,
how it was fixed
P Original linker (Who Finishes the Job?)
M Added in Mohamed's extension
O Added in this thesis